

THE SOLAR FLUX INFLUENCE ON QUIET TIME CONDUCTANCES IN THE AURORAL IONOSPHERE

Jøran Moen* and Asgeir Brekke**

Abstract. Hall and Pedersen conductances derived from EISCAT Common Program 1 (CP-1) data during half a solar cycle are presented. The data set consists of 9 quiet summer days with 10.7-cm solar flux (S_a) ranging from solar minimum ($S_a = 74$) to solar maximum ($S_a = 248$) conditions. The solar flux and the solar zenith angle (χ) dependence of Hall (Σ_H) and Pedersen (Σ_P) conductances are well represented by

$$\Sigma_H = S_a^{0.53} (0.81 \cos \chi + 0.54 \cos^{1/2} \chi)$$

$$\Sigma_P = S_a^{0.49} (0.34 \cos \chi + 0.93 \cos^{1/2} \chi)$$

respectively. The present results are in good agreement with the theoretical model calculations by Rasmussen et al. [1988] for solar minimum conditions. For higher solar activity levels, however, our model gives consistently higher conductance values than older models.

Introduction

From well-known classical equations [Chapman, 1956], the electrical conductivity in a plasma is linearly proportional to the electron density. At the E- and F₁-altitudes (below say 180 km) the high latitude electron density profile is in general mainly controlled by the solar ultraviolet (EUV) radiation and auroral particle precipitation. At higher altitudes, however, transport processes become increasingly important.

According to the work by Moen and Brekke [1990] pertinent to solar minimum conditions, is the contribution to the Hall conductance above 170 km negligible, and less than 30% to the Pedersen conductance above this altitude. Thus, in a sunlit ionosphere, when precipitating particles are absent, variations in the Hall and Pedersen conductances are expected to reflect the variations in the electron density due to a varying solar insulation caused by variations in the solar zenith angle and the solar activity.

The current paper is devoted to establishing empirical formulas for the Hall and Pedersen conductances including effects of the solar flux represented by the S_a index as well as the solar zenith angle (χ) by using EISCAT incoherent scatter data of the electron density profiles.

The EISCAT Common Program One (CP-1) is performed by the UHF (933 MHz) system with the antenna directed approximately along the magnetic field line in Tromsø (elevation 77.54° and azimuth 182.57°) [Rishbeth and Williams, 1985]. The electron density profile derived from the CP-1 experiment with proper spatial as well as temporal resolution has been extensively used in auroral conductivity and conductance studies (cf. review paper by Brekke and Moen [1992] and references therein). Similar studies of conductances have previously been made by using the Chatanika [cf. Robinson and

Vondrak, 1984] and the Saint-Santin [Senior, 1980] incoherent scatter radars.

The EISCAT radar equipment has been operated routinely since 1985. The EISCAT CP-1 data base, covering the five years period (1985–1990) provides the first opportunity to study how the conductances in the ionosphere above Tromsø depend upon the solar flux.

In section 2 we briefly describe the computational model used to derive the conductances and justify the selection of the data ensemble. In section 3 we describe the stepwise attempt used to relate the Hall and Pedersen conductances to variations in the solar zenith angle (χ) as well as the solar flux index (S_a). In section 4 a comparison of our empirical results with earlier works is made. The paper is concluded in section 5.

Analysis Method – Data Selection

In this work we use an improved version of the computational model for conductances as outlined by Brekke and Hall [1988]. In their work, they made use of the raw electron density profiles observed by EISCAT CP-1 experiments to compute the Hall and Pedersen conductances. According to the incoherent scatter radar technique, the raw electron density is assumed to be directly proportional to the total backscattered power received from the plasma volume investigated [Baron, 1977].

The raw electron densities are, however, related to the true electron densities by a correction factor given by [Farley, 1966; Rishbeth and Williams, 1985]:

$$N_e \propto N_{raw} (1 + T_e/T_i + \alpha^2) (1 + \alpha^2) \quad (1)$$

accounting for unequal ion (T_i) and electron (T_e) temperatures as well as Debye length effects (α). At F-region heights the electron temperature is significantly higher than the ion temperature. Then the raw electron density becomes an underestimate of the true electron density. Such raw electron density profiles have been used in several conductivity studies based on incoherent scatter radar measurements [Robinson and Vondrak, 1984; Brekke and Hall, 1988; de la Beaujardière et al., 1991]. As has been pointed out by the authors, the underestimate is usually less than 25%, since at these relevant F-region heights the electron temperature is generally not larger than two times the ion temperature and the contribution to the conductances from these regions are usually small. However, the relative contribution from the F-layer to the Hall and Pedersen conductances is larger for solar maximum than minimum conditions due to the increase in the F-layer electron densities toward maximum conditions. As a consequence the underestimate in conductances derived from raw electron density profiles is expected to be larger during periods of solar maximum than solar minimum conditions.

In the present work we have inferred a corrected (true) electron density profile by merging together the multipulse signals from the altitude region 87 km to 180 km with a height resolution of 2.7 km and the long pulse signals from the altitude region 180 km to 330 km with a height resolution of 22.5 km. The rather crude spatial resolution above 180 km is adequate for the present study,

*Department of Physics, University of Oslo, Norway

**The Auroral Observatory, University of Tromsø, Norway

TABLE 1.

Day	S_a	A_p	A_H	B_H	A_P	B_P
May 27, 1986	77	7	7.50	5.23	0.73	8.96
July 15, 1986	74	4	10.47	3.30	3.97	6.38
May 12, 1987	86	2	11.87	2.96	3.90	7.77
June 16, 1987	78	14	8.59	5.10	1.15	8.71
May 03, 1988	113	7	4.05	10.30	2.94	9.55
Aug 01, 1989	199	6	14.43	11.15	2.80	15.28
June 05, 1990	148	5	15.75	5.44	4.29	11.48
July 02, 1990	248	6	5.44	15.29	8.18	9.06
July 30, 1990	180	12	14.54	5.51	5.83	11.43

because at F-region heights the altitude variations in the electron densities are small over height distances smaller than say 25 km. The numerical integration is performed by applying the trapezoidal rule. We also apply the electron and ion temperatures derived simultaneously from EISCAT observations as input in the models for the ion neutral and electron neutral collision frequencies adopted from Schunk and Walker [1973] and Schunk and Nagy [1978], respectively. This is in contrast to the assumption of thermal equilibrium; $T_e = T_i = T_n$ where T_n is the neutral temperature, as done by Brekke and Hall [1988], an improvement found, however, to be of minor importance.

In order to select an appropriate data ensemble from the entire EISCAT CP-1 data base, we exercise the following criteria:

- Minimum solar zenith angle during the day $\chi_{min} \leq 50^\circ$.
- Maximum solar zenith angle $\chi_{max} \leq 80^\circ$.
- Quiet conditions, no particle precipitation.

The criterium a) is met by using summer time data only. χ_{max} is set to 80° in order to assure that the dominant ionization source during quiet conditions is the solar irradiation.

At sunset ($\chi \geq 90^\circ$) the electron density profile is maintained at some level. At night time for quiet periods, processes such as EUV radiation from starlight, resonantly scattered solar radiation and radiative recombination become increasingly important [Strobel et al., 1980].

Attempts have been made to establish quiet night time empirical models of the Hall and Pedersen conductances based on EISCAT CP-1 programme data. Since Tromsø is situated well within the active auroral oval between 1900 and 0500 local time on the average, this has so far not been a very successful exercise.

By using the criteria above, the remaining data ensemble consists of 9 quiet summer days presented in Table 1 along with the solar flux (S_a) – the solar radio flux index in units of $10^{-22} \text{ W m}^{-2} \text{ Hz}^{-1}$ as observed at 10.7 cm and adjusted to 1 AU – and magnetic (A_p) indices. Our data base is selected over half a solar cycle with the solar flux index varying between 74 and 248 units.

The Daily Variation of the Hall and Pedersen Conductances

Typical quiet time Hall and Pedersen conductances as derived by the EISCAT CP-1 program are shown in Figure 1 as function of universal time in Tromsø (Local Time (LT) = UT + 1 hr) for a couple of experiments. Smooth variations are seen in both the Hall (Σ_H) and the Pedersen (Σ_P) conductances in daytime, responding

to the regular variation in the solar irradiation. At night time, however, sporadic increases are seen in both components, but relatively stronger in Σ_H than in Σ_P as the Tromsø site crosses through the auroral oval.

Results such as these have been used extensively in the past to estimate the effect of the solar zenith angle variations on the Hall and Pedersen conductances (see Brekke and Moen [1992] for review). During quiet daytime summer conditions these parameters are shown to be well represented by the empirical solar zenith angle relationship

$$\Sigma_{H,P}(\chi) = A_{H,P} \cos \chi + B_{H,P} \cos^{1/2} \chi \quad (2)$$

where $A_{H,P}$ and $B_{H,P}$ are constants to be determined by a least squares fit of the data set to this equation [Brekke and Hall, 1988]. In Table 1 we have listed the coefficients $A_{H,P}$ and $B_{H,P}$ obtained by the method of least squares fitting to (2) of the computed Hall and Pedersen conductances for the 9 days listed in Table 1. The values for the different coefficients are in reasonable agreement with the corresponding values derived from raw electron density profiles by Brekke and Hall [1988].

Barth et al. [1990] have shown that there is a high level of correlation between the solar radio flux at 10.7 cm as indicated by the S_a index and the flux of the solar Lyman-alpha emission. Since all the days in our data set except May 27 and July 15 in 1986 correspond to the maximum phase as defined by Barth et al. [1990], we expect the solar radio flux index for the present data set to be a good indicator of the Lyman-alpha emission flux. With regard to other parts of the solar UV and EUV spectrum which are more effective agents for E-region ionization, the correlation with the solar radio

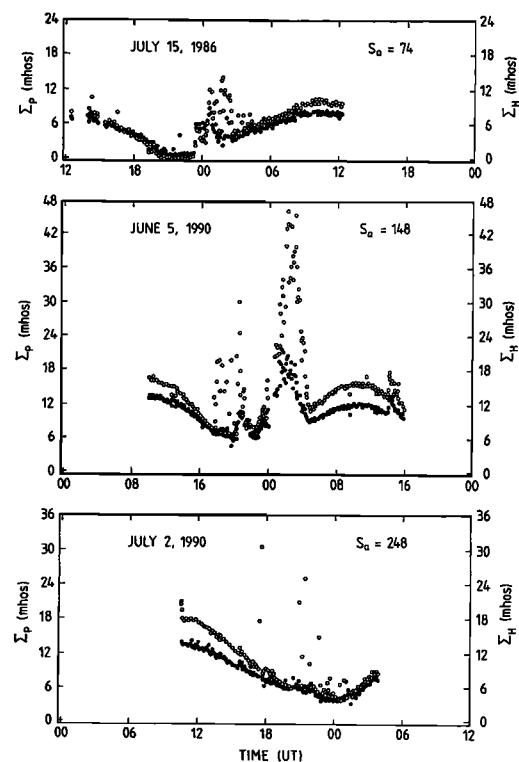


Fig. 1. Hall and Pedersen conductances versus universal time (UT = LT - 1 hr) as obtained by the EISCAT CP-1 programme explained in the text. Upper panel for July 15, 1986. Middle panel for June 5, 1990. Lower panel for July 2, 1990.

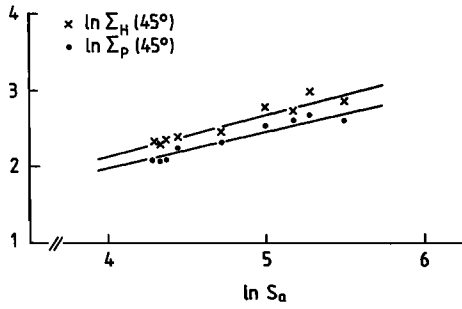


Fig. 2. Hall and Pedersen conductances derived for $\chi = 45^\circ$ versus the solar flux index S_a (in $10^{-22} \text{ W m}^{-2} \text{ Hz}^{-1}$ as observed at 10.7 cm wavelength and adjusted to 1 A.U.). The Σ values are according to the empirical formulas given in (2) in the text with the respective coefficients $A_{H,P}$ and $B_{H,P}$ as listed in Table 1.

flux is less apparent [Hall et al., 1985]. It therefore remains to be shown whether the solar radio flux at 10.7 cm is a representative indicator for the solar radiation flux ionizing the ionospheric E-region. In lack of a more appropriate index, however, we follow a similar approach as originally introduced by Robinson and Vondrak [1984] and search for a relationship between the coefficients A and B in (2) and the solar flux index (S_a).

We therefore propose that for a fixed solar zenith angle ($\chi = 45^\circ$) the conductances can be expressed by

$$\Sigma_{H,P}(45, S_a) = S_a^{\gamma_{H,P}} \Sigma_{H,P}(45) \quad (3)$$

where

$$\Sigma_{H,P}(45) = A_{H,P}/S_a^{\gamma_{H,P}} \cos 45 + B_{H,P}/S_a^{\gamma_{H,P}} \cos^{1/2} 45 \quad (4)$$

where γ_H and γ_P are constants to be determined empirically. Compared to the work by Robinson and Vondrak [1984] we find these coefficients given by: $\gamma_{H,P} = 1/2$, $A_{H,P} = 0$ and $B_{H,P} = 1$.

Figure 2 shows the values derived of the Hall and Pedersen conductances at a zenith angle of 45° for each of the 9 days listed in Table 1 as function of the corresponding S_a values. Both axes are on logarithmic scales. Also indicated in the figure are the results of a least squares fitting approach represented by

$$\ln \Sigma_{H,P} = \gamma_{H,P} \ln S_a + C_{H,P} \quad (5)$$

where C_H and C_P are constants. The slope of the lines are represented by γ_H or γ_P . The appropriate values for these coefficients according to Figure 2 are

$$\gamma_H = 0.53$$

$$\gamma_P = 0.45$$

respectively, in very good agreement with the values assumed by Robinson and Vondrak [1984]. The γ -values obtained were found to be independent of the χ -value chosen in (3), which indicates that S_a and χ are independent variables and thus separable. By now taking the appropriate γ -values and inserting for S_a and $A_{H,P}$ and $B_{H,P}$ in (4) the ratios $A_{H,P}/S_a^{\gamma_{H,P}}$ and $B_{H,P}/S_a^{\gamma_{H,P}}$ are derived and listed in Table 2. These ratios are now chosen to be representing the solar zenith angle effect on Σ_H and Σ_P . According to the mean values derived in Table 2 the empirical formulas for the quiet summer daytime Hall and Pedersen conductances including both

TABLE 2.

Day	S_a	$A_H/S_a^{\gamma_H}$	$B_H/S_a^{\gamma_H}$	$A_P/S_a^{\gamma_P}$	$B_P/S_a^{\gamma_P}$
May 27, 1986	77	0.75	0.52	0.09	1.07
July 15, 1986	74	1.07	0.34	0.48	0.77
May 12, 1987	86	1.12	0.28	0.44	0.88
June 16, 1987	78	0.85	0.51	0.14	1.03
May 03, 1988	113	0.33	0.84	0.29	0.94
Aug 01, 1989	199	0.87	0.67	0.21	1.14
June 05, 1990	148	1.11	0.38	0.37	0.99
July 02, 1990	248	0.29	0.82	0.55	0.61
July 30, 1990	180	0.93	0.35	0.45	0.77
Average		0.81	0.54	0.34	0.93

$\gamma_H = 0.53$ and $\gamma_P = 0.49$ as derived from Figure 2 and (5).

S_a - The solar radio flux index in units of $10^{-22} \text{ W m}^{-2} \text{ Hz}^{-1}$ as observed at 10.7 cm and adjusted to 1 AU.

solar zenith angle and solar flux dependences can now be summarized by

$$\begin{aligned} \Sigma_H(\chi, S_a) &= S_a^{0.53} (0.81 \cos \chi + 0.54 \cos^{1/2} \chi) \\ \Sigma_P(\chi, S_a) &= S_a^{0.49} (0.34 \cos \chi + 0.93 \cos^{1/2} \chi) \end{aligned} \quad (6)$$

Comparison with Other Works

As already mentioned the γ -values are in close agreement (within $\pm 6\%$) with the values applied by Robinson and Vondrak [1984] and based on physical arguments concerning an assumed linear relationship between S_a and the solar flux actually responsible for the ionization in the lower ionosphere. If this assumption is valid the produced electron density in photochemical equilibrium will be proportional to $\sqrt{S_a}$ if dissociative recombination is the dominant loss term at the heights where N_e is contributing to the conductances. The good agreement obtained here to the work by Robinson and Vondrak [1984] therefore supports the assumption made by them concerning the solar flux relationship.

In Figure 3 the Pedersen conductance is shown versus the solar zenith angle according to (6) for 4 different days. For the lowest solar index the empirical model overestimates the measured values by about 0.5 mhos independent of the zenith angle. In the medium range solar activity index the empirical model underestimates the measured values by close to 1 mho, and finally for the highest solar flux index ($S_a \approx 250$) there is a discrepancy of almost 2 mhos between the model and the measured values. The model is overestimating the measurements. When we compare our results in more detail with the work of Robinson and Vondrak [1984] and Rasmussen et al. [1988] the present model gives higher values for all zenith angles than either of the two others. The difference is increasing towards higher S_a values and can be as large as 4 mhos for the lowest solar zenith angle. For the very low solar flux value ($S_a = 70$) a very good agreement is obtained between this work and Rasmussen et al. [1988].

The differences obtained between these results and previous studies are not clearly understood. There are indications, however, that Rasmussen et al. [1988] underestimated the electron densities in the E-region in their model calculations compared to the expectations based on relevant Chatanika observations.

Robinson and Vondrak [1984] underestimated the conductances up to 25% by using raw electron density profiles in their calculations. For the lowest solar zenith angle in Figure 3 their model results are also 20–30%

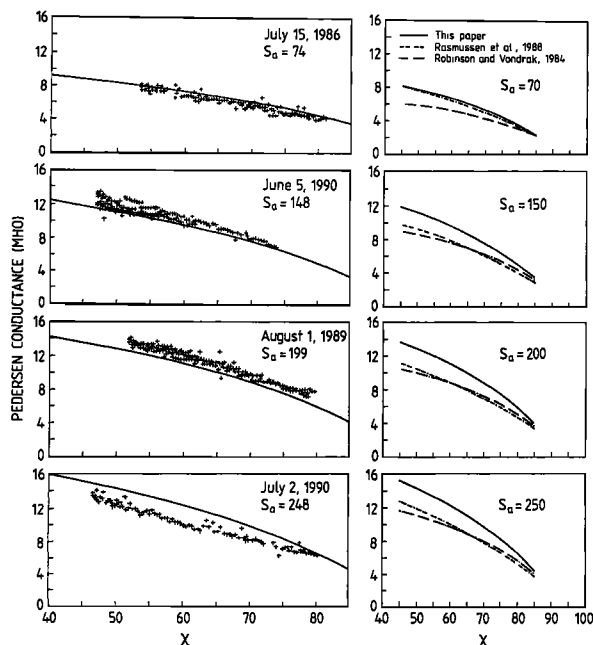


Fig. 3. The Pedersen conductance versus the solar zenith angle for 4 individual days with different solar flux indices. The observations are compared with the empirical model as derived in the text.

In the right-hand side panels the present model is compared to the models derived by Robinson and Vondrak [1984] and Rasmussen et al. [1988] for the respective days.

lower than ours, suggesting that previous conductance models have been somewhat underestimated.

Conclusion

Hall and Pedersen conductances derived in the summer quiet daytime auroral ionosphere are very well-behaved and show a strong relationship to the variation in the solar zenith angle. This rather systematic behaviour has been further analysed in a search for a solar irradiation relationship. It is found that for the range of the solar flux index (S_a) studied between 70 and 250 ($\times 10^{-22} \text{ W m}^{-2} \text{ Hz}^{-1}$) the Pedersen and Hall conductances increase very closely as the square root of S_a . While the empirical model for the Pedersen conductance so derived based on 9 data sets overestimates the observed value for the highest solar flux values, a fairly good agreement is obtained for the lowest flux values. For the medium range solar fluxes ($150 < S_a < 200$) a slight underestimation of the observed values results from the empirical model. Although our data base is limited, by comparing the present result with earlier work by Robinson and Vondrak [1984] and Rasmussen et al. [1988], older conductance models seem to have been underestimations.

Acknowledgement: We are indebted to Mrs. Liv Larssen for all assistance in preparing this manuscript. We also wish to thank the EISCAT staff for their valuable collaboration. EISCAT is being jointly funded by the Science and Engineering Research Council (U.K.), Centre National de la Recherche Scientifique (France), Max-Planck Gesellschaft (F.R.G.), Suomen Akatemia (Finland), Norges Allmennvitenskapelige Forskningsråd (Norway) and Naturvetenskapliga Forskningsrådet (Sweden). This project has been supported by grants 427.91/014 and 427.91/025 from Norges Allmennviten-

skapelige Forskningsråd and by Prof. L.L. Cogger, the University of Calgary, Calgary, Canada.

References

- Baron, M. J., The Chatanika Radar System, in *Radar Probing of the Auroral Plasma*, edited by A. Brekke, pp. 103–142, Universitetsforlaget, Oslo, 1977.
- Barth, C. A., W. Kent Tobiska, G. J. Rottman, and O. R. White, Comparison of 10.7 cm radio flux with the solar Lyman alpha flux, *Geophys. Res. Lett.*, **17**, 571–574, 1990.
- de la Beaujardière, O., R. Johnson, and V. B. Wickwar, Ground-Based Measurements of Joule Heating Rates, in *Auroral Physics*, edited by C. I. Meng, M. J. Rycroft, and L. A. Frank, pp. 439–448, Cambridge Univ. Press, 1991.
- Brekke, A., and C. Hall, Auroral Ionospheric Quiet Summer Time Conductances, *Ann. Geophys.*, **6**, (4), 361–376, 1988.
- Brekke, A., and J. Moen, Observations of high latitude ionospheric conductances – a review. To be submitted to *J. Atmos. Terr. Phys.*, 1992.
- Chapman, S., The electrical conductivity of the ionosphere: A review, *Nuovo Cimento*, **4**, 1385–1412, 1956.
- Farley, D. T., A theory of incoherent scattering of radio waves by plasma, *J. Geophys. Res.*, **71**, 4091–4098, 1966.
- Hall, L. A., L. J. Heroux, and H. E. Hinteregger, Solar ultraviolet irradiance. Chap. 2. *Handbook of Geophysics and the Space Environment*, ed. A. S. Jura, Air Force Geophysics Laboratory, United States Air Force, 1985.
- Moen, J., and A. Brekke, On the importance of ion composition to conductivities in the auroral ionosphere, *J. Geophys. Res.*, **92**, 10687–10693, 1990.
- Rasmussen, C. E., R. W. Schunk, and V. B. Wickwar, A photochemical equilibrium model for ionospheric conductivity, *J. Geophys. Res.*, **93**, 9831–9840, 1988.
- Rishbeth, H., and P. J. S. Williams, The EISCAT ionospheric radar: the system and its early results, *Q. J. Roy. Astron. Soc.*, **26**, 478–512, 1985.
- Robinson, R. M., and R. R. Vondrak, Measurements of E region ionization and conductivity produced by solar illumination at high latitudes, *J. Geophys. Res.*, **89**, 3915–3956, 1984.
- Schunk, R. W. and A. F. Nagy, Electron temperature in the F region of the ionosphere: theory and observations, *Rev. Geophys. Space Phys.*, **16**, 355–399, 1978.
- Schunk, R. W. and J. C. G. Walker, Theoretical ion densities in the lower ionosphere, *Planet. Space Sci.*, **21**, 1875–1896, 1973.
- Senior, C., Les conductivités ionosphériques et leur rôle dans la convection magnétosphérique: Une étude expérimentale et théorique, Diplôme de docteur de 3^e cycle, L'Univ. Pierre et Marie Curie, Paris, 1980.
- Strobel, D. F., C. B. Opal, and R. R. Meier, Photoionization rates in the night-time E- and F-region ionosphere, *Planet. Space Sci.*, **28**, 1027–1033, 1980.

Jørn Moen, Department of Physics, P. O. Box 1048, N-0315 Oslo, Norway.

Asgeir Brekke, The Auroral Observatory, University of Tromsø, N-9037 Tromsø, Norway

(Received: February 4, 1992;
revised: June 5, 1992;
accepted: August 24, 1992.)